Rishik Thammisetty

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Summary

Electrical and Electronics Engineering undergraduate at NIT Andhra Pradesh with hands-on experience in digital design, VLSI, RTL development, static timing analysis, and FPGA prototyping. Experienced in Verilog-based digital systems and ML-assisted EDA workflows for timing and CMOS circuit analysis. Strong foundation in digital logic, circuit design, timing analysis, and hardware-oriented machine learning.

Education

National Institute of Technology Andhra Pradesh — Tadepalligudem, India
B.Tech. in Electrical and Electronics Engineering; CGPA: 8.35/10 (Current) — ***Aug 2024 – Present***
Minor Degree in AI and Robotics — ***Aug 2025 – Present***

Technical Skills

Programming: C, C++, Python, Verilog, MATLAB
Digital Design: RTL Design, Digital Logic, Pipelining, Hazard Detection, Data Forwarding, Cache Design
VLSI / EDA: Static Timing Analysis, Timing Graphs, Gate-Level Netlists, CMOS Circuit Modeling, RTL-to-GDS
Hardware / FPGA: Xilinx Vivado, FPGA Prototyping, STM32, ESP32, Raspberry Pi
ML / EDA: PyTorch, Scikit-learn, NumPy, Pandas, Physics-Informed ML, ML-assisted EDA

Professional Experience

IEEE EMBS Chapter, Indian Institute of Technology (IIT) Kharagpur — **Aug 2026 – Present**
Intern – Medical Image Analysis
- Developing and evaluating deep learning models for computational pathology, focusing on algorithm optimization and quantitative image-quality metrics.

Indian Institute of Technology (IIT) Bhilai — Paper in Progress — **May 2026 – Aug 2026**
Summer Research Intern | *Github*
- Explored computationally efficient architectures and hardware acceleration strategies targeting **FPGA implementation** for real-time deep learning inference.

e-Yantra Robotics Club, NIT Andhra Pradesh — **Aug 2025 – Aug 2026**
Technical Volunteer
- Organized hands-on workshops on drone technology, teaching students the fundamentals of **embedded systems**, drone architecture, and **firmware** configuration.
- Mentored peers on hardware-software integration, guiding them through microcontroller programming (e.g., STM32) and sensor interfacing for robotics applications.

Academic Projects

32-bit Pipelined RISC-V Processor with Cache | *GitHub* — **Aug 2026 – Present**
- Designing a 32-bit RISC-V processor in Verilog RTL with a five-stage pipeline: IF, ID, EX, MEM, and WB.
- Implementing hazard detection and data forwarding, with instruction/data caches and branch prediction for improved execution efficiency.
- Studying and applying the RTL-to-GDS flow including synthesis, timing analysis, physical design, and FPGA prototyping.
- Performed pre-silicon verification by developing Verilog testbenches and executing behavioral simulations in Xilinx Vivado to validate pipeline functionality and functional correctness.

- **Machine Learning-Assisted Static Timing Analysis** | *GitHub* — **May 2026 – Present**
  - Developed an ML-assisted STA framework combining gate-level netlist parsing, timing-graph construction, physics-aware features, and deep-learning-based delay prediction.
  - Built an end-to-end EDA pipeline for timing propagation, critical-path identification, and bottleneck analysis using BLIF netlists.
  - Achieved 0.82% timing error against SPICE on C17 and ∼49× speedup on a 432-gate, 216-bit ripple-carry adder using FT-Transformer and physics-informed models.
- **Physics-Informed ML for CMOS Gate Delay Prediction** | *GitHub* — **Jan 2026 – Apr 2026**
  - Generated SPICE-based 45nm CMOS gate-delay data across transistor sizing and process-voltage-temperature (PVT) conditions.
  - Developed physics-informed neural networks for CMOS delay prediction while enforcing physically meaningful behavior across operating conditions.
  - Compared MLP and FT-Transformer models and used SHAP to analyze the influence of transistor and operating parameters.

Certifications & Coursework

- **Certifications**
  - **NPTEL**: Design and Analysis of VLSI Subsystems (Elite Certified)
  - **NPTEL**: VLSI Design Flow: RTL to GDS (On-going)
  - **Qualcomm Academy**: Hands-on Model Development and AI Fundamentals
  - **NVIDIA**: Building Transformer-Based Natural Language Processing Applications